

PHILOSOPHY 130: PHILOSOPHY OF SPACE AND TIME
ASSIGNMENT 4: GALILEAN SPACETIME

Due: April 28, in class

Name: _____

Reading: Maudlin, Ch. 3 (“Eliminating Unobservable Structure”); Geroch, Chs. 3–4.

Please draw and write your answers in the space provided. Note: there will not be an assignment distributed on April 30; instead we will distribute the exam questions for the exam on May 5.

1. Geroch on parallel worldlines. In Chapters 3–4, Geroch emphasizes that building up Galilean spacetime requires a notion of *parallel straight worldlines* — a way of judging that two inertial observers, at different places, are moving in “the same way.” Consider the following scenario. Two astronauts, Alice and Bob, drift through deep space, each carrying an accelerometer that reads zero. They are far apart and exchange no signals. (Each is moving inertially, as far as they can determine locally.)

- (a) A pure *Machian* spacetime (cf. Maudlin, Ch. 2) provides only the relative distances between bodies at each moment and the elapsed times between events. In 2–3 sentences, explain why such a spacetime does *not* supply enough structure to answer the question: “Are Alice and Bob moving parallel to one another, or are their worldlines tilted with respect to each other?”

- (b) Now suppose Alice and Bob are in fact moving so that the distance between them is constant. According to a full-fledged *relationalist* who admits only spatial relations between bodies and elapsed times, is there a further fact of the matter about whether they are *both* inertial, or whether (say) they are both accelerating in circles around a common center while maintaining constant separation? Explain in 2–3 sentences, connecting your answer to Newton’s discussion.

- (c) Geroch’s Galilean spacetime settles the question in (b) affirmatively, because it includes a distinguished class of **straight worldlines**. In 2–3 sentences, explain what additional structure (beyond Machian distances and times) this amounts to, and why, according to Maudlin’s *mismatch principle*, Newtonian physics requires at least this much.

Suppose a cosmologist proposes the following new theory, *Theory T*. Theory T keeps Newton’s three laws of motion unchanged, but adds a single new postulate: *every isolated body experiences an additional “cosmic drag” force directly proportional to its absolute velocity through space.*

- (a) Under Theory T, would a Galilean boost (setting the whole universe in uniform motion) produce an observationally equivalent situation, or a genuinely different one? Justify your answer in 2–3 sentences.
- (b) Does Theory T satisfy Maudlin’s mismatch principle with respect to full Newtonian spacetime? In 2–3 sentences, argue either that it does, or that it does not, being careful to say exactly which symmetries are being compared.